

HOUGHTON COUNTY AP, MI, USA

WMO: 727440

Lat: 47.169N

Lon: 88.489W

Elev: 334

StdP: 97.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14858

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.4	-24.8	0.4	-20.7	-22.3	0.5	-18.1	14.5	-6.1	12.8	-6.4	4.7	310	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	28.9	21.1	27.2	19.9	25.5	19.0	22.4	27.3	21.2	25.5	20.0	24.1	4.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	16.0	25.5	19.5	14.8	23.9	18.4	13.9	22.7	67.3	27.3	62.8	25.5	58.9	24.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.8	9.1	DB	-24.2	32.9	3.3	1.7	-26.6	34.1	-28.6	35.0	-30.4	36.0	-32.9	37.2	(4)
(5)			WB	-24.5	24.6	3.3	1.2	-26.9	25.4	-28.8	26.1	-30.6	26.8	-33.0	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-8.7	-8.1	-3.5	3.1	10.2	15.6	18.6	18.0	14.1	7.2	0.4	-5.4	(6)	
	DBStd	10.91	5.45	5.87	6.25	4.97	4.82	4.13	3.54	3.31	4.60	4.71	4.99	5.03	(7)	
	HDD10.0	2679	579	508	420	214	57	4	0	0	14	116	291	477	(8)	
	HDD18.3	4921	837	741	677	457	257	102	41	47	142	346	539	736	(9)	
	CDD10.0	925	0	0	2	7	62	172	268	249	136	28	2	0	(10)	
	CDD18.3	124	0	0	0	0	4	19	49	37	14	1	0	0	(11)	
	CDH23.3	943	0	0	1	2	53	169	371	249	91	7	0	0	(12)	
	CDH26.7	206	0	0	0	0	13	47	79	49	18	1	0	0	(13)	
(14) Wind	WSAvg	4.3	4.8	4.8	4.5	4.5	4.3	3.9	3.9	3.8	3.9	4.5	4.6	4.6	(14)	
(15) Precipitation	PrecAvg	806	90	49	50	48	67	69	72	69	80	66	65	73	(15)	
	PrecMax	1073	216	130	190	146	145	168	157	164	168	144	136	185	(16)	
	PrecMin	438	6	2	5	8	12	16	15	6	28	26	18	2	(17)	
	PrecStd	160	48	32	37	26	32	36	34	36	36	32	30	42	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	7.2	15.4	21.8	28.3	31.3	31.2	30.4	29.1	23.6	16.4	7.1	(19)	
		MCWB	2.0	4.1	11.1	12.9	18.5	22.1	22.6	21.8	21.1	16.3	10.6	3.9	(20)	
	2%	DB	2.0	4.0	10.3	17.1	24.1	27.5	28.7	27.9	26.0	19.9	12.2	4.2	(21)	
		MCWB	0.7	1.7	6.5	9.5	16.0	19.3	21.3	20.8	19.3	14.4	8.9	2.5	(22)	
	5%	DB	0.7	2.0	7.2	13.7	21.2	24.9	27.1	26.1	23.6	17.1	9.5	2.7	(23)	
		MCWB	-0.2	0.6	4.4	7.9	13.9	18.2	20.2	19.5	18.3	12.9	6.5	1.5	(24)	
	10%	DB	-0.9	0.0	4.7	10.8	18.4	22.9	25.4	24.3	21.4	14.3	7.3	1.4	(25)	
		MCWB	-1.8	-1.1	2.3	6.0	12.5	16.9	19.2	18.5	17.0	11.2	5.3	0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	4.4	12.4	14.2	19.9	23.8	24.1	23.6	22.0	17.8	12.0	5.0	(27)	
		MCDB	3.7	6.6	14.9	19.6	25.0	30.0	29.1	27.9	27.0	21.8	14.8	6.2	(28)	
	2%	WB	0.9	2.2	7.1	10.5	17.2	21.2	22.5	21.9	20.5	15.4	9.5	2.9	(29)	
		MCDB	1.8	3.7	9.5	15.6	22.5	25.8	27.3	26.6	24.4	18.6	12.2	3.7	(30)	
	5%	WB	-0.2	0.4	4.6	8.1	15.2	19.5	21.2	20.6	19.1	13.4	6.9	1.6	(31)	
		MCDB	0.6	1.8	7.0	13.1	19.8	23.5	25.6	24.5	22.6	16.2	8.9	2.7	(32)	
	10%	WB	-1.8	-1.1	2.3	6.3	13.0	17.9	20.0	19.5	17.6	11.5	5.4	0.4	(33)	
		MCDB	-0.9	0.0	4.4	10.3	17.5	21.5	24.2	23.3	20.7	14.0	7.2	1.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	7.4	8.4	8.6	10.6	10.9	10.5	10.1	9.4	7.5	5.9	5.6	(35)	
		MCDBR	6.8	9.2	10.4	13.9	14.8	13.5	12.3	12.2	12.1	11.7	9.7	7.0	(36)	
	5% WB	MCWBR	6.2	7.4	7.6	8.8	8.3	7.7	6.8	6.5	6.6	7.5	6.8	6.0	(37)	
		MCWBR	6.5	8.4	9.8	12.8	13.6	12.4	11.2	10.7	10.8	10.3	8.9	6.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.259	0.278	0.302	0.338	0.367	0.374	0.384	0.366	0.356	0.321	0.291	0.260	(40)	
		taud	2.451	2.378	2.389	2.377	2.386	2.381	2.408	2.456	2.468	2.535	2.534	2.486	(41)	
	Ebn at Noon		850	899	925	917	897	890	875	878	855	841	804	808	(42)	
		Edh at Noon	64	86	100	112	115	117	112	103	93	74	61	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.03	1.99	3.45	4.63	5.50	5.99	6.12	5.21	3.81	2.26	1.15	0.76	(44)	
	RadStd		0.10	0.24	0.33	0.51	0.40	0.41	0.27	0.34	0.21	0.27	0.17	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page